



Overview

Question

Test Cases

Code

Instructions for Extended Match QuestionsUse this slide to understand how to answer an EMQ**Instructions:**

1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. [Please click here to view a sample template file for answering.](#)
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

Consider the following matrices: $R_1 = \begin{bmatrix} 10 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, $R_2 = \begin{bmatrix} 1001 \\ 0101 \\ 0010 \end{bmatrix}$, $R_3 = \begin{bmatrix} 10 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $R_4 = \begin{bmatrix} 10 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

List of Options:

1. R_1 is a row echelon form of AA .
2. R_1 is the reduced row echelon form of AA
3. R_2 is the reduced row echelon form of AA
4. R_3 is the reduced row echelon form of AA
5. R_4 is the reduced row echelon form of AA
6. $\text{Nullspace}(A) = \{(-t, -t, 0, t) \mid t \in \mathbb{R}\}$
7. $\text{Nullspace}(A) = \{(-t, \frac{t}{2}, t, 0) \mid t \in \mathbb{R}\}$
8. $\text{Nullspace}(A) = \{(-\frac{5}{2}t_1 - t_2, -\frac{3}{4}t_1 - t_2, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
9. $\text{nullity}(A) = 1$
10. $\text{nullity}(A) = 2$
11. Basis of $\text{Nullspace}(A) = \{(-\frac{5}{2}, -\frac{3}{4}, 1, 0), (-1, -1, 0, 1)\}$
12. Basis of $\text{Nullspace}(A) = \{(1, -\frac{1}{2}, -1, 0)\}$
13. Basis of $\text{Nullspace}(A) = \{(1, 1, 0, -1)\}$

List of Questions:

Q1 Consider the matrix $A = \begin{bmatrix} 20 & 52 \\ 12 & 43 \\ 20 & 72 \end{bmatrix}$, which of the above options are true for AA ?

Q2 Consider the matrix $A = \begin{bmatrix} 2 & 0 & 5 \\ 1 & 2 & 4 \\ 3 & 2 & 9 \end{bmatrix}$ $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 2 & 2 \\ 5 & 4 & 9 \end{bmatrix}$, which of the above options are true for AA ?

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$